

1. What is Google Colab?

Google Colab (Google Colaboratory) is a **free cloud-based platform** that allows users to write and execute **Python code** using a notebook interface.

It is built on top of **Jupyter Notebooks** and provides access to **Google's computing resources**, including CPUs, GPUs, and TPUs.

2. Why Google Colab is Used

Google Colab is widely used because it removes many common barriers faced by beginners and professionals.

Main Reasons:

- No need to install Python or libraries
- Runs completely in the browser
- Free access to powerful hardware
- Easy sharing and collaboration
- Ideal for data science, machine learning, and data engineering practice

3. How to Set Up Google Colab (Step-by-Step)

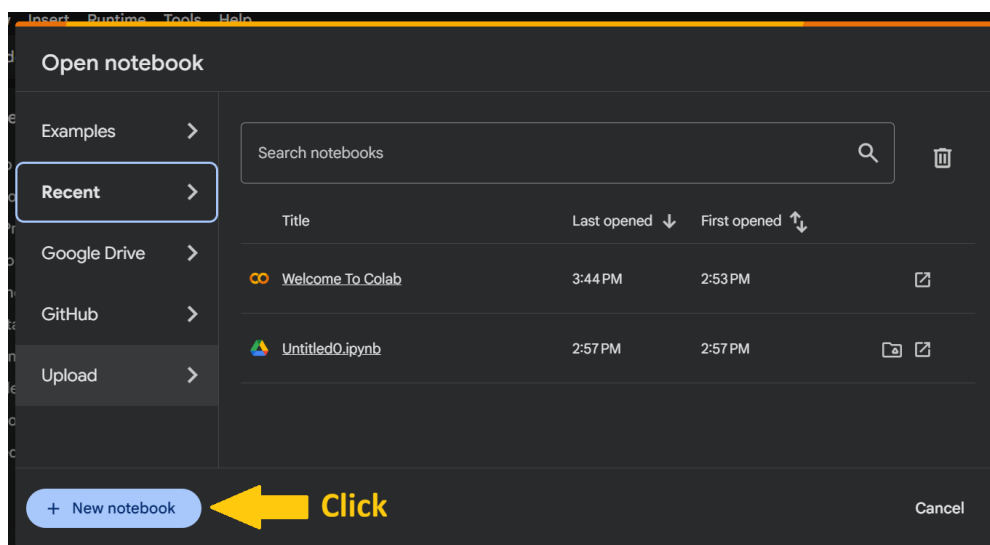
Step 1: Open Google Colab

Open your browser and go to:

<https://colab.research.google.com>

Step 2: Create a New Notebook

On the welcome screen:



A new Python notebook will open.

Step 3: Understand the Notebook Interface

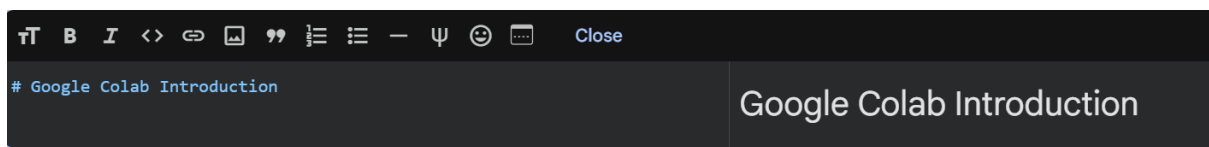
A notebook contains two main types of cells:

Code Cell : Used to write and execute Python code.



Text Cell : Used to write explanations using Markdown.

Example:



Step 4: Connect to Runtime

At the top-right corner:

- Click **Connect**

This starts a runtime where your code will execute.

Step 5: Select Runtime Type (Optional)

To change hardware:

- Go to **Runtime** → **Change runtime type**
- Choose:
 - CPU (default)
 - GPU
 - TPU
- Click **Save**

For beginners, CPU is sufficient.

Google Colab Keyboard Shortcuts

(Code Cells & Markdown Cells)

Google Colab has **two modes**, just like Jupyter:

- **Command Mode** (cell-level actions)
- **Edit Mode** (typing inside a cell)

1. Modes in Google Colab

⚡ Command Mode

- Used to manage cells
- Cell border is **blue**
- Press Esc to enter command mode

⚡ Edit Mode

- Used to type code or text
- Cell border is **green**
- Press Enter to enter edit mode

2. Cell Execution Shortcuts

Shortcut	Action
Shift + Enter	Run cell & move to next
Ctrl + Enter	Run cell & stay
Alt + Enter	Run cell & insert new cell below

3. Code Cell & Markdown Cell Shortcuts

Convert Cell Type (Command Mode)

Shortcut	Action
Y	Convert to Code Cell
M	Convert to Markdown Cell

4. Cell Creation Shortcuts (Command Mode)

Shortcut	Action
A	Insert cell above
B	Insert cell below

5. Cell Deletion & Movement

Shortcut	Action
D D	Delete selected cell
Ctrl + M + ↑	Move cell up
Ctrl + M + ↓	Move cell down

6. Editing Shortcuts (Edit Mode)

Shortcut	Action
Ctrl + /	Comment / Uncomment
Ctrl + D	Delete line
Tab	Autocomplete
Shift + Tab	Show docstring / help

7. Markdown Formatting Shortcuts (Edit Mode)

Shortcut	Result
#	Heading 1
##	Heading 2
###	Heading 3
text	Bold
<i>*text*</i>	Italic

8. Navigation Shortcuts

Shortcut	Action
↑ / ↓	Navigate between cells
Ctrl + M + L	Toggle line numbers
Ctrl + F	Find
Ctrl + H	Find & Replace

9. Runtime Shortcuts

Shortcut	Action
Ctrl + M + .	Interrupt execution
Ctrl + M + I	Restart runtime
Ctrl + M + B	Clear outputs